

GR&R Study Report

Gauge Repeatability & Reproducibility — AIAG MSA 4th Edition

Equipment / Gage:	Zeiss Contura CMM SN-4471
Study Performed By:	QE Lab - Example C
Report Date:	2026-05-10
Study Design:	3 Operators, AIAG Crossed GR&R
Regulatory Reference:	21 CFR 820.72 AIAG MSA 4th Ed.

Variation Components Summary

Component	Study Variation	% of TV	Threshold
EV — Equipment (Repeatability)	0.0058	6.2%	—
AV — Appraiser (Reproducibility)	0.0049	5.3%	—
GR&R — Combined	0.0076	8.2%	≤ 10%: Accept ≤ 30%: Marginal
PV — Part Variation	0.0923	99.7%	—
TV — Total Variation	0.0926	100.0%	—
ndc — Number of Distinct Categories	17.2	—	≥ 5 required

Intermediate Calculation Values

Parameter	Symbol	Value	Formula
Grand Mean	X-bar-bar	14.9932	Mean of all measurements
Avg Range (overall)	R-bar-bar	0.0019	Mean of operator R-bars
Operator Mean Spread	X-diff	0.0019	max(X-bar op) - min(X-bar op)
Part Range	Rp	0.0570	max(part avg) - min(part avg)

Per-Operator Breakdown

Operator	Grand Mean (X-bar)	Avg Range (R-bar)
Alice	14.9932	0.0018
Bob	14.9941	0.0021
Carol	14.9922	0.0018

Acceptance Determination

[PASS] ACCEPTABLE

%GR&R = 8.2% | ndc = 17.2

The measurement system is ACCEPTABLE per AIAG MSA 4th Edition criteria. %GR&R; is at or below 10%, indicating the gage contributes minimal variation relative to total process variation. This instrument may be used for product acceptance decisions. Continue routine calibration per your control plan.

ndc = 17.2 — ADEQUATE: The gage can resolve 5 or more distinct categories of part variation.

AIAG Constants Applied — K1=? (10 trials), K2=2.7 (3 operators), K3 per part count. Study variation = 5.15σ (99% of normal distribution).

Reference: AIAG Measurement Systems Analysis Reference Manual, 4th Edition (2010). Regulatory basis: 21 CFR Part 820.72 — Inspection, Measuring, and Test Equipment (FDA Quality System Regulation). This report is a controlled quality record. Retain per applicable document control procedures.